

Abstract

Pais-type nonequilibrium electromagnetic architectures form a controversial but experimentally instructive class of proposed advanced-propulsion and energy systems. These architectures typically combine high-frequency electromagnetic excitation, accelerated vibration or spin, charged boundary surfaces, resonant cavities, plasma or dielectric media, and claims of vacuum-state modification or inertial-mass reduction. While such claims remain unverified and are not accepted as established physics, the systems themselves occupy an experimentally important regime characterized by strong nonlinear coupling, transient electromagnetic stress redistribution, and multiple competing conventional force pathways.

This work develops a comparative experimental framework for evaluating PTNEA systems through force-decomposition analysis, artifact-discrimination logic, Maxwell-stress considerations, Poynting-vector energy transport, and EVRT-compatible residual-response normalization using the effective susceptibility parameter, χ_{eff} . Multiple architecture families are examined, including high-energy electromagnetic field generators, inertial-mass reduction concepts, resonant microwave cavities, plasma compression fusion devices, high-frequency gravitational-wave systems, and rotational Lorentz-force configurations.

Rather than asserting evidence for propulsion, antigravity, or spacetime engineering, this paper establishes a falsifiable methodology for determining whether residual effects remain after conventional electromagnetic, thermal, mechanical, plasma, acoustic, and environmental contributions are independently bounded below measurement sensitivity thresholds. The resulting framework is intended to support disciplined experimental evaluation of highly coupled nonequilibrium electromagnetic systems.

Artifact Discrimination and Residual-Force Metrology in Pais-Type Nonequilibrium Electromagnetic Architectures

Erick S. Sangalang

Independent Researcher, Coherent Nonequilibrium Electromagnetic Systems Series

May 6, 2026

Scope and positioning. This paper does not claim validation of antigravity, reactionless propulsion, or spacetime-engineering mechanisms. Instead, it develops a comparative methodological framework for evaluating Pais-type nonequilibrium electromagnetic architectures through conservation-law constraints, artifact discrimination, and residual-force normalization.

No new fundamental physics is claimed. The present framework does not introduce a validated extension to electrodynamics, gravitation, or quantum field theory. Instead, it establishes a phenomenological and experimentally falsifiable methodology for evaluating whether reproducible residual responses remain after conventional electromagnetic, thermal, plasma, acoustic, vibrational, and mechanical mechanisms are independently bounded.

Terminology. For compactness, this paper refers to Pais-type nonequilibrium electromagnetic architectures as PTNEA systems. The acronym is used only as a descriptive classification label and does not imply validation of any claimed propulsion, inertial-modification, or vacuum-engineering mechanism.

1 Introduction

PTNEA systems are a family of proposed electromagnetic systems that invoke high-frequency excitation, charged boundary surfaces, accelerated vibration or spin, resonant cavities, plasma or dielectric media, and far-from-equilibrium drive conditions. The most prominent examples include high-energy electromagnetic field generator concepts, inertial-mass reduction devices, hybrid aerospace-undersea craft architectures, high-frequency gravitational-wave generators, plasma compression fusion systems, and proposed room-temperature superconducting systems.

The central scientific challenge is not simply whether such systems can generate complex electromagnetic states. Rather, the relevant question is whether any measured macroscopic force or anomalous response remains after conventional electromagnetic, thermal, plasma, acoustic, vibrational, and mechanical pathways have been independently bounded. This paper therefore treats PTNEA systems as stress tests for experimental force metrology rather than as established propulsion mechanisms.

Regardless of their ultimate physical interpretation, PTNEA systems provide unusually demanding environments for precision force metrology because multiple nonlinear electromagnetic, thermal, plasma, and mechanical channels coexist simultaneously.

Claims of anomalous electromagnetic propulsion and inertial modification have appeared periodically across several experimental families, including asymmetric capacitor systems, resonant microwave cavities, rotating gyroscope experiments, high-voltage dielectric systems, and plasma-assisted electromagnetic devices. These reports differ greatly in evidentiary quality, but they share a

common methodological challenge: small apparent force signals can be produced by many ordinary mechanisms unless the experimental environment is tightly constrained.

2 Pais-Type Architecture Families

PTNEA systems may be grouped into several architecture families. These categories are useful because each family emphasizes a different combination of electromagnetic, mechanical, plasma, and material-response pathways.

Table 1: PTNEA architecture families and EVRT-compatible interpretation.

Architecture	Claimed mechanism	EVRT-compatible interpretation	Dominant risks
HEEMFG	High EM energy flux and vacuum polarization	Transient EM stress-energy architecture	RF leakage, heating, vibration
IMRD / HAUC	Inertial-mass reduction	Residual-force claim requiring bounds	Mechanical rectification, plasma effects
HFGW generator	EM-to-GW conversion via Gertsenshtein effect	Extreme field-coupling claim	Magnetic recoil, RF coupling
RTSC	Vibration-induced superconductivity	Nonequilibrium material-response claim	Thermal drift, contact resistance
PCFD	Plasma compression fusion	Nonlinear plasma/EM confinement system	Plasma pressure, magnetic coupling
Rotational EM systems	Lorentz and gyroscopic asymmetry	Momentum-redistribution claim	Bearing friction, imbalance, magnetic coupling

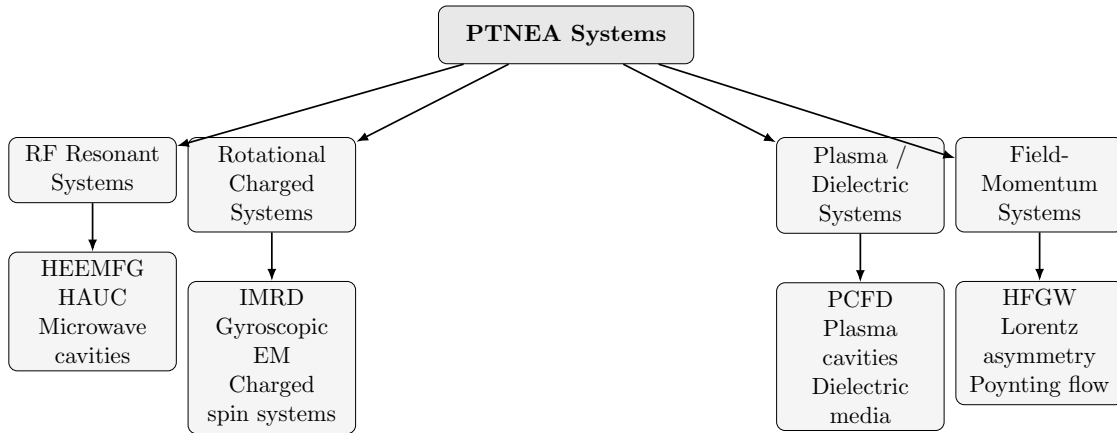


Figure 1: Taxonomy of Pais-type nonequilibrium electromagnetic architectures. Systems are grouped by dominant coupling pathways rather than by claimed propulsion interpretation.

3 Baseline Physics and Force Decomposition

The evaluation of PTNEA systems should begin with established electrodynamics and mechanics. Candidate mechanisms include Maxwell stress, Lorentz forces, Poynting-vector momentum flow, RF heating, plasma pressure, acoustic excitation, dielectric charging, magnetic coupling, and mechanical recoil.

The measured force may be decomposed as

$$F_{\text{measured}} = F_{\text{EM}} + F_{\text{Lorentz}} + F_{\text{plasma}} + F_{\text{thermal}} + F_{\text{mechanical}} + F_{\text{RF}} + F_{\text{environmental}} + F_{\text{residual}}. \quad (1)$$

Only the final term should be interpreted as a candidate residual after the known terms have been independently bounded below the measurement threshold.

4 Conservation-Law Constraints on Pais-Type Architectures

Any proposed residual-force interpretation associated with PTNEA systems must remain consistent with conservation of momentum and conservation of energy. In classical electrodynamics, electromagnetic momentum transport is governed by Maxwell stress tensors and Poynting-vector energy flow. For a closed system, internal Lorentz forces should cancel unless momentum is transferred externally through radiation, plasma interaction, environmental coupling, or measurement artifacts.

A useful continuum expression is

$$\frac{d\mathbf{P}}{dt} = \oint \mathbf{T} \cdot d\mathbf{A} + \frac{\partial}{\partial t} \int \mathbf{g}_{\text{EM}} dV, \quad (2)$$

where $\mathbf{T}$ is the Maxwell stress tensor and $\mathbf{g}_{\text{EM}}$ represents electromagnetic momentum density. Consequently, any experimentally reported net thrust or residual force must ultimately be reconciled with momentum transport, radiation balance, or externally coupled interactions.

4.1 Null Results as Experimental Constraints

Strong null results remain scientifically valuable within the EVRT framework because they place quantitative upper bounds on allowable nonequilibrium electromagnetic coupling strengths. Failure to observe reproducible residual forces under controlled vacuum, polarity-reversal, geometry-reversal, dummy-load, and thermal-isolation conditions constrains the parameter space available to Pais-type interpretations.

4.2 Uncertainty and Systematic Error

Residual-force interpretation is fundamentally limited by calibration uncertainty, environmental coupling, thermal drift, vibration sensitivity, RF leakage, cable forces, magnetic coupling, pressure dependence, and systematic measurement bias. Consequently, any inferred residual should be reported with both statistical uncertainty and a systematic-error budget. In the present framework, an apparent residual is not considered meaningful unless the combined uncertainty of conventional channels is smaller than the reported residual signal.

5 EVRT-Compatible Residual Normalization

To compare widely different electromagnetic architectures on a common phenomenological basis, an effective residual-response normalization parameter may be defined as

$$\chi_{\text{eff}} = \frac{F_{\text{residual}}}{u_{\text{EM}}V}, \quad (3)$$

where F_{residual} denotes the experimentally inferred residual force after artifact subtraction, u_{EM} represents characteristic electromagnetic energy density, and V is the active interaction volume.

The characteristic electromagnetic energy density used for normalization may be estimated, at leading order, by

$$u_{\text{EM}} = \frac{1}{2} \left(\epsilon_0 E^2 + \frac{B^2}{\mu_0} \right), \quad (4)$$

where E and B are representative electric and magnetic field amplitudes in the active region. This expression does not imply that the full field distribution is uniform; rather, it supplies a controlled scaling quantity for comparing devices of different sizes, field strengths, and geometries.

The parameter χ_{eff} is not interpreted here as a fundamental constant. Instead, it serves as a comparative phenomenological metric allowing constraint-based comparison between different nonequilibrium electromagnetic architectures.

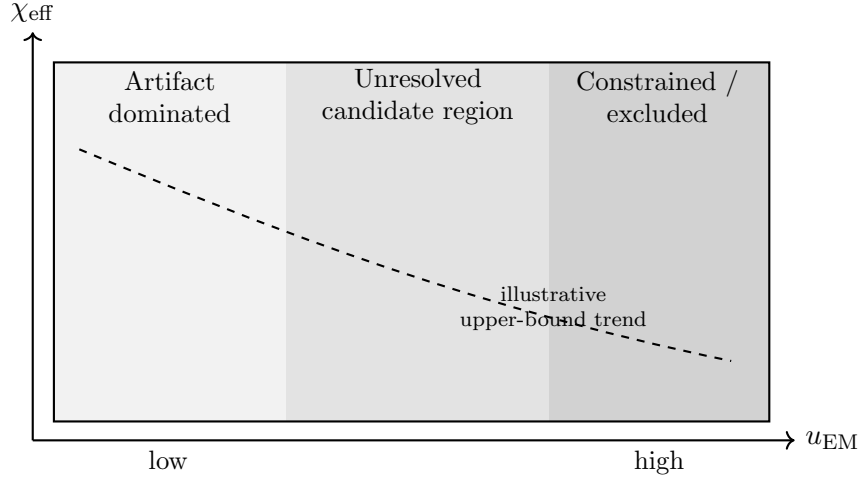


Figure 2: Conceptual parameter-space map for comparing inferred χ_{eff} bounds across nonequilibrium electromagnetic architectures. The diagram is schematic and represents interpretation regimes rather than measured universal constants.

5.1 Illustrative Numerical Bound

For a representative tabletop resonant cavity, take an active volume

$$V = 1.5 \times 10^{-4} \text{ m}^3,$$

a characteristic electric field amplitude

$$E = 1.0 \times 10^5 \text{ V m}^{-1},$$

and assume the magnetic contribution is subdominant for this simple estimate. Equation (4) then gives

$$u_{\text{EM}} \simeq \frac{1}{2} \epsilon_0 E^2 \simeq 4.4 \times 10^{-2} \text{ J m}^{-3}.$$

If the post-subtraction residual-force sensitivity is

$$F_{\text{residual}} \leq 1.0 \times 10^{-6} \text{ N},$$

then Eq. (3) gives the illustrative upper bound

$$\chi_{\text{eff}} \lesssim \frac{1.0 \times 10^{-6}}{(4.4 \times 10^{-2})(1.5 \times 10^{-4})} \approx 1.5 \times 10^{-1} \text{ m}^{-1}.$$

This value is not a universal constant. It is a device-specific phenomenological bound determined by geometry, field strength, active volume, and residual-force sensitivity. Its usefulness lies in allowing different experiments to be compared using the same normalization convention.

6 Relationship to Existing Null-Result Literature

Historically, many reported anomalous-force systems become less persuasive as experimental controls improve. The spinning-gyroscope literature is a useful example: the original Hayasaka–Takeuchi result motivated interest in possible rotation-dependent weight anomalies, but subsequent controlled measurements reported null results or placed strong limits on the effect. Within the present framework, this history is not treated as a failure of experimental inquiry. Rather, it illustrates why increasingly strict controls are essential when small force signals may be contaminated by bearing friction, vibration, air currents, magnetic coupling, or scale drift.

The implication for PTNEA systems is direct. A claimed residual force should not be evaluated only by its nominal magnitude, but by whether it survives systematic rejection of conventional mechanisms. Conversely, a null result under well-characterized conditions remains valuable because it bounds the allowed residual coupling and reduces the viable parameter space for nonequilibrium electromagnetic interpretations.

7 Levels of Interpretation

To maintain experimental rigor, Pais-type interpretations may be separated into several evidentiary levels.

8 Artifact Pathway Matrix

PTNEA systems are especially vulnerable to overlapping artifact channels because they combine strong electromagnetic excitation, vibration, heating, plasma, and mechanical support interactions.

9 Experimental Protocol Framework

9.1 Minimum Experimental Requirements

A credible experimental evaluation of PTNEA systems should minimally include:

- vacuum-compatible testing environments;

Table 2: Interpretation hierarchy for PTNEA systems.

Interpretation level	Status
Conventional electromagnetic and mechanical artifacts	Established
Nonlinear electromechanical or plasma coupling	Plausible and testable
Residual-force candidate after artifact subtraction	Unverified
Vacuum polarization and spacetime coupling	Speculative
Inertial-mass reduction or gravitational modification	Highly speculative

Table 3: Artifact pathway matrix for PTNEA systems.

Artifact channel	Expected signature	Scaling behavior	Diagnostic test
Cable Lorentz force	Directional force aligned with wiring	Current-dependent	Cable rerouting / dummy load
RF heating	Delayed drift	Power and duty-cycle dependent	Thermal lag measurement
Plasma pressure	Strong gas dependence	Pressure and composition dependent	Vacuum sweep
Mechanical vibration	Phase-correlated force	Frequency-dependent	Accelerometer isolation
EHD / ion wind	Atmosphere-sensitive thrust	Voltage and pressure dependent	High-vacuum operation
Magnetic recoil	Orientation-dependent force	Magnetic-field dependent	Magnetic shielding / reversal
Dielectric charging	Slow relaxation and hysteresis	Voltage history dependent	Discharge and polarity cycling

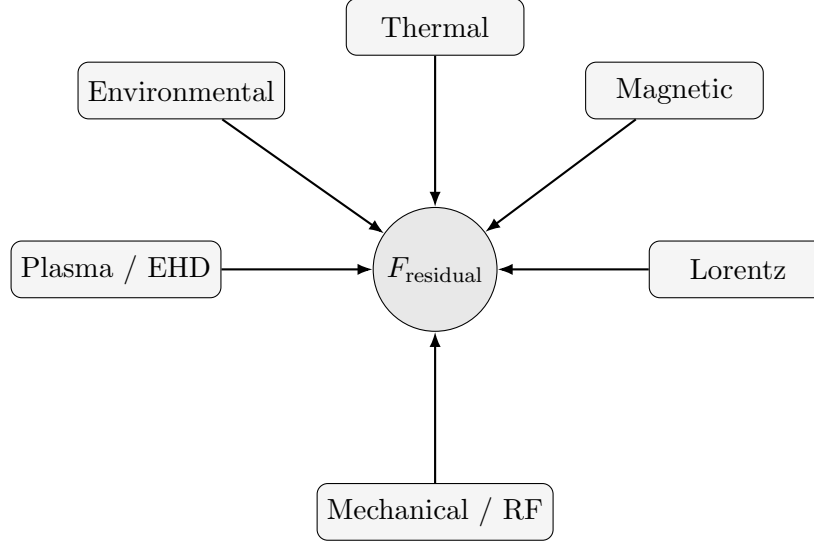


Figure 3: Force-pathway compass illustrating why PTNEA systems require multi-axis artifact discrimination. Candidate residual behavior must be separated from thermal, Lorentz, plasma, RF, magnetic, environmental, and mechanical pathways.

- polarity and geometry reversal tests;
- thermal characterization and lag analysis;
- dummy-load substitution measurements;
- vibration isolation and accelerometer monitoring;
- magnetic shielding and cable-force analysis;
- independent replication where possible;
- automated or blind data acquisition protocols.

These controls are necessary because PTNEA systems combine multiple strongly coupled electromagnetic, thermal, acoustic, plasma, and mechanical pathways that may mimic residual-force signatures.

9.2 Experimental Prediction Matrix

10 Why Pais-Type Systems Are Difficult to Evaluate

PTNEA systems occupy a challenging experimental regime because they combine transient high-frequency electromagnetic fields, charged boundaries, resonant cavity behavior, plasma effects, thermal gradients, and mechanical vibration in highly nonlinear configurations. Small systematic errors may therefore produce force-like signatures that are difficult to separate from candidate residual phenomena.

The resulting signal environment often involves RF leakage and cable Lorentz forces, thermal expansion and drift, electrohydrodynamic flow, plasma pressure asymmetry, mechanical rectification, magnetic coupling to nearby structures, and phase-dependent nonlinear oscillatory behavior.

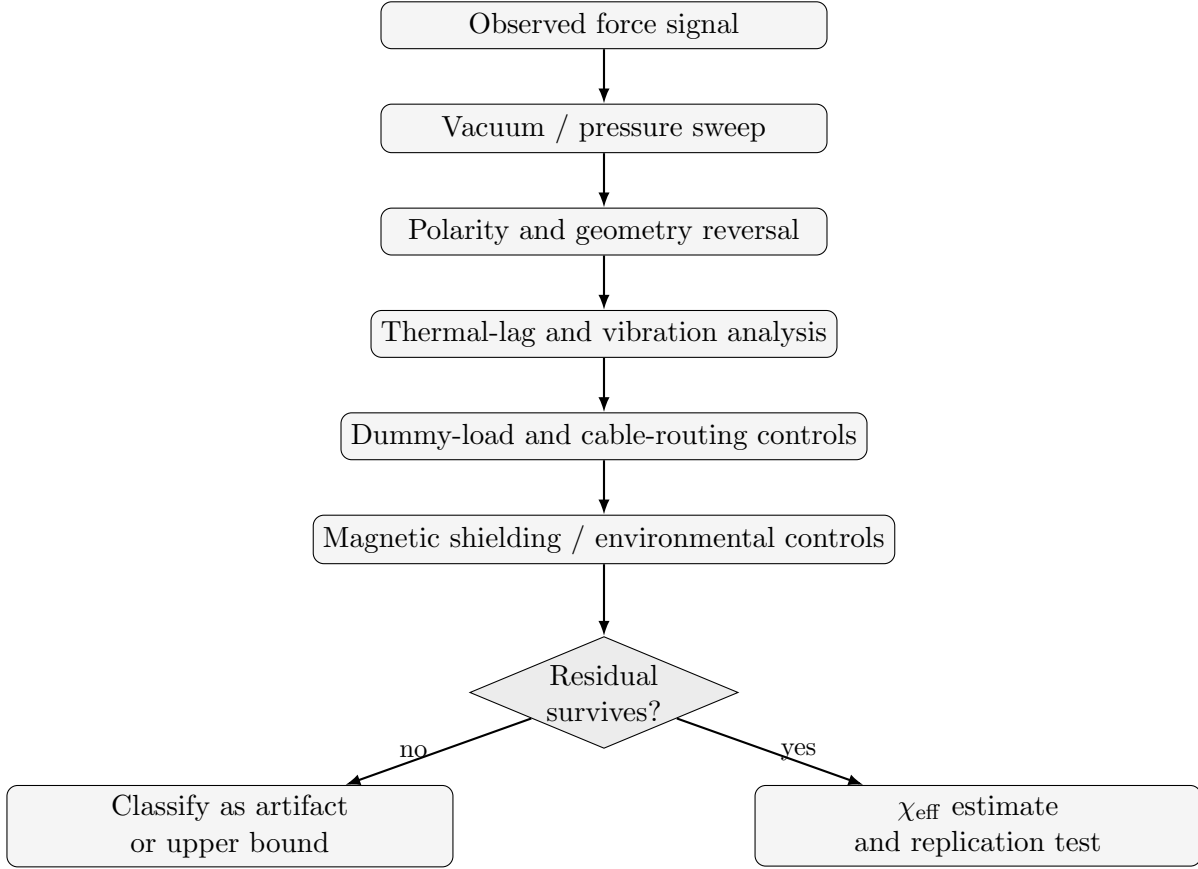


Figure 4: Artifact-discrimination flowchart for PTNEA force experiments.

Table 4: Experimental prediction matrix distinguishing conventional artifacts from candidate residual behavior.

Observable	Conventional expectation	Residual-response expectation
Pressure dependence	Strong for EHD/plasma pathways	Weak or systematically bounded
Polarity reversal	Sign or magnitude changes	Geometry/topology dependent
Thermal lag	Delayed response	Prompt response after artifact subtraction
Frequency sweep	Resonance peaks and mechanical modes	Reproducible nonthermal scaling
Geometry inversion	Artifact axis may follow supports/wires	Residual axis should follow field topology
Dummy-load substitution	Signal persists if caused by wiring/heating	Signal should vanish or change predictably

Consequently, disciplined artifact-discrimination methodologies remain central to any scientifically credible interpretation of Pais-type residual-force claims.

11 Illustrative Residual Bound Categories

Table 5: Illustrative residual-force bound categories for EVRT-compatible comparison.

Experiment class	Observable scale	Conventional limiting channels	Residual status
Asymmetric capacitors	mg–g claims	EHD, corona, electrostatic coupling	artifact-sensitive
RF resonators	μN claims	thermal drift, RF leakage, cable forces	weakly constrained
Rotating charged systems	mg claims	bearing friction, vibration, charge leakage	not independently replicated
Plasma cavities	variable	plasma pressure, thermal gradients	artifact dominated
Pais-type HEEMFG concepts	not independently verified	RF, thermal, mechanical, plasma pathways	unbounded

12 Worked Example: Hypothetical Resonant Pais-Type Test Article

As an illustrative example, consider a compact resonant test article consisting of a charged asymmetric cavity driven at radio frequency inside a vacuum-compatible torsion-balance enclosure. The measured signal is assumed to be a small angular deflection converted into an apparent force F_{measured} through independent calibration of the torsion constant.

The analysis would proceed as follows. First, the raw force signal is decomposed according to Eq. (1). Cable Lorentz forces are tested by rerouting feed lines and substituting dummy loads. Thermal contributions are bounded by monitoring cavity temperature, heat-sink gradients, and time lag between RF power and apparent force. Plasma and EHD contributions are tested through pressure sweeps from atmosphere to high vacuum. Mechanical vibration is bounded using accelerometers and off-resonance drive conditions. Magnetic recoil is tested by rotating the apparatus or adding shielding.

If a residual remains after these controls, it may be normalized using Eq. (3). For example, if the active volume is V , the characteristic energy density is estimated by Eq. (4), and the residual after artifact subtraction is F_{residual} , then χ_{eff} provides a comparative bound rather than a claim of universal coupling. If the residual vanishes under dummy-load substitution or pressure reduction, the result is classified as artifact dominated. If no reproducible residual remains above the noise floor, the experiment yields an upper bound on χ_{eff} for that architecture.

13 Discussion

The value of PTNEA systems lies not in their current evidentiary status, but in their usefulness as stress tests for nonequilibrium electromagnetic force metrology. These systems collect many of the hardest experimental problems into a single architecture: strong RF fields, plasma effects, charged surfaces, resonant structures, vibration, thermal drift, and claims of residual momentum exchange.

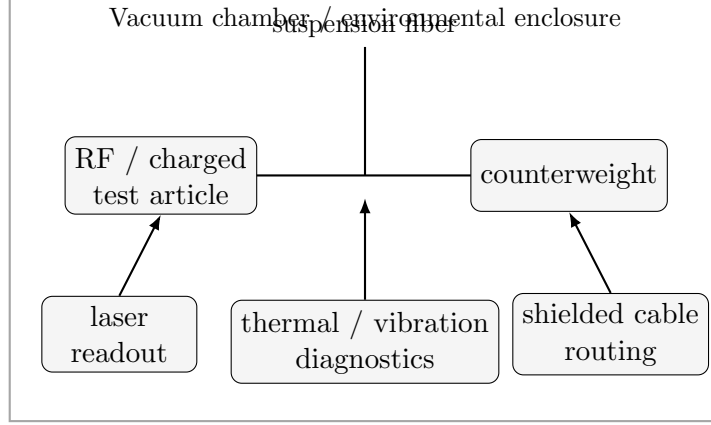


Figure 5: Schematic torsion-balance test layout for a Pais-type resonant electromagnetic architecture. The figure emphasizes diagnostic channels rather than a specific propulsion claim.

Within the EVRT-compatible interpretation, a Pais-type residual claim becomes meaningful only if it survives artifact subtraction, conservation-law accounting, and reproducible scaling analysis. If no residual is observed, the resulting null result remains valuable because it constrains the parameter space of allowed nonequilibrium electromagnetic coupling.

14 What Would Count as Strong Evidence?

A compelling PTNEA residual-force claim would require more than a single force trace or apparent scale deflection. At minimum, a strong result should demonstrate:

- reproducibility across repeated runs and independent apparatus configurations;
- persistence under high-vacuum conditions where EHD and plasma effects are strongly suppressed;
- predictable behavior under polarity, orientation, and geometry reversal;
- consistency with thermal, RF, vibration, and magnetic diagnostics;
- disappearance or predictable modification under dummy-load substitution;
- scaling with field energy density, resonant state, or geometry in a way that is not explained by known artifacts;
- independent replication by a separate experimental group.

Within this framework, the most persuasive evidence would not be a large isolated signal, but a reproducible residual with a stable χ_{eff} scaling behavior after conventional mechanisms have been quantitatively excluded.

15 Limitations

The framework developed here is intentionally conservative and phenomenological. It does not provide a microscopic derivation for a new vacuum-response mechanism, does not establish a first-principles propulsion model, and does not claim that PTNEA systems have produced reproducible

anomalous forces. The parameter χ_{eff} is a comparative residual-response metric, not a universal material constant or fundamental coupling.

A further limitation is that the framework depends on experimental implementation quality. Poor thermal control, inadequate cable isolation, insufficient vacuum testing, or incomplete vibration monitoring can make residual interpretation ambiguous. For this reason, the proposed methodology should be viewed as a constraint-analysis structure rather than as evidence for any specific exotic mechanism.

16 Conclusion

Pais-type nonequilibrium electromagnetic architectures provide a useful but controversial class of systems for developing disciplined experimental interpretation methods. By applying force decomposition, Maxwell-stress constraints, artifact pathway analysis, and χ_{eff} normalization, these architectures can be reframed from speculative propulsion claims into testable experimental categories.

The framework developed here does not validate inertial-mass reduction, antigravity, or spacetime-engineering claims. Instead, it establishes a conservative methodology for determining whether any reproducible residual response remains after known electromagnetic, plasma, thermal, mechanical, and environmental mechanisms have been independently bounded. In this way, PTNEA systems become valuable stress tests for precision nonequilibrium electromagnetic force metrology. Even strong null results would remain scientifically valuable by establishing progressively tighter bounds on candidate nonequilibrium electromagnetic coupling mechanisms.

A Recommended Experimental Reporting Standards

To improve reproducibility and comparison across PTNEA investigations, experimental reports should include:

- vacuum pressure and gas composition;
- temperature monitoring and thermal lag measurements;
- cable-routing geometry and current paths;
- vibration-monitoring instrumentation;
- magnetic shielding and environmental isolation procedures;
- dummy-load and polarity-reversal test results;
- calibration methodology and uncertainty propagation;
- data-acquisition automation and blind-analysis procedures where possible.

The goal of these reporting standards is not to validate residual-force claims a priori, but to improve consistency, reproducibility, and artifact discrimination across highly coupled nonequilibrium electromagnetic experiments.

References

- [1] S. C. Pais, “The High Energy Electromagnetic Field Generator,” *International Journal of Space Science and Engineering* **3**, 312–317 (2015).
- [2] S. C. Pais, “Conditional Possibility of Spacecraft Propulsion at Superluminal Speeds,” *International Journal of Space Science and Engineering* **3**, 89–92 (2015).
- [3] S. C. Pais, “A Hybrid Craft Using an Inertial Mass Modification Device,” AIAA SPACE Forum, AIAA 2017-5343 (2017).
- [4] S. C. Pais, “High Frequency Gravitational Waves – Induced Propulsion,” SAE Technical Paper 2017-01-2040 (2017).
- [5] S. C. Pais, “Room Temperature Superconducting System for Use on a Hybrid Aerospace-Undersea Craft,” AIAA SciTech Forum, AIAA 2019-0869 (2019).
- [6] S. C. Pais, “The Plasma Compression Fusion Device—Enabling Nuclear Fusion Ignition,” *IEEE Transactions on Plasma Science* **47**, 5119–5126 (2019).
- [7] J. D. Jackson, *Classical Electrodynamics*, 3rd ed. (Wiley, 1998).
- [8] L. D. Landau and E. M. Lifshitz, *Electrodynamics of Continuous Media*, 2nd ed. (Pergamon, 1984).
- [9] P. W. Milonni, *The Quantum Vacuum: An Introduction to Quantum Electrodynamics* (Academic Press, 1994).
- [10] J. Schwinger, “On Gauge Invariance and Vacuum Polarization,” *Physical Review* **82**, 664–679 (1951), doi:10.1103/PhysRev.82.664.
- [11] A. Taflov and S. C. Hagness, *Computational Electrodynamics: The Finite-Difference Time-Domain Method* (Artech House, 2005).
- [12] H. Hayasaka and S. Takeuchi, “Anomalous Weight Reduction on a Gyroscope’s Right Rotations around the Vertical Axis on the Earth,” *Physical Review Letters* **63**, 2701–2704 (1989).
- [13] J. M. Nitschke and P. A. Wilmarth, “Null Result for the Weight Change of a Spinning Gyroscope,” *Physical Review Letters* **64**, 2115–2116 (1990).
- [14] J. E. Faller, W. J. Hollander, P. G. Nelson, and M. P. McHugh, “Gyroscope-Weighing Experiment with a Null Result,” *Physical Review Letters* **64**, 825–826 (1990).
- [15] M. Tajmar and I. Lorincz, “Measurement of Anomalous Forces in Rotating Systems,” *Measurement* **73**, 453–461 (2015).
- [16] H. White, P. March, J. Lawrence, J. Vera, A. Sylvester, D. Brady, and P. Bailey, “Measurement of Impulsive Thrust from a Closed Radio-Frequency Cavity in Vacuum,” *Journal of Propulsion and Power* **33**, 830–841 (2017), doi:10.2514/1.B36120.
- [17] M. E. McCulloch, “Can the EMDrive Be Explained by Quantised Inertia?” *EPL* **111**, 60005 (2015), doi:10.1209/0295-5075/111/60005.